Forensic analysis report

This is an analysis of what could be shown from the file. It is NOT a certificate of authenticity.

File	H2_signed_then_edited.pdf
SHA-256	`85b46f65caa8d128581ed364e5c7c009d7dff2227613a25d3f512c7aa79d88d0`
Analysed	2026-11-06T20:00:00Z by Doc Forensics Engine 0.1.0 (ruleset 2026.11-r1)

REPORT CATEGORY

Confirmed change

A change is demonstrated by a trusted original or by cryptographic evidence.

Why this category

* Signature "Signature1": the document was changed after it was signed - The signature is cryptographically valid for the earlier revision, which anchors that version as what "Grace Banda" signed. The later revisions contain different content, so the file now differs from the signed version. The signed version can be reconstructed and compared.
* 3 further indicator(s) are also listed for review.

No probability or confidence score is given: none of these checks has been statistically calibrated.

1. DOCUMENT OVERVIEW

Size	18720 bytes
SHA-512	`cb700ff967dc8fc993fe4ecf3d7e1e2fc1bb4823c97aa9ba5663ef075ce6d1173fc841c09c1c382a2b522d121f927a5fcba4f66baeffd020e0e538ff2a6f01f2`
SHA-1	`30d3789c4b9b33ff9a5df2f016f2bfba82c7d6dc`
MD5	`3d1387b9cf9effecde9bd950a15cb299`
PDF version	1.3
Pages	1
Revisions	3 (2 incremental update(s))
Creator	anonymous
Producer	ReportLab PDF Library - (opensource); pyHanko 0.38.1
Created	2000-01-01 00:00:00 UTC
Modified	2026-11-06 19:25:29 UTC
Version 0	bytes 0-7836, SHA-256 f836f642806d539077bb419a9284cae3fbce2ba8f4dda7c63a7ff8ad0dcc2076
Version 1	bytes 7836-16626, SHA-256 379801f6cb753a372bd8098580609432c7bfdf489073d3aa88c7033d2045c58a
Version 2	bytes 16626-18720, SHA-256 85b46f65caa8d128581ed364e5c7c009d7dff2227613a25d3f512c7aa79d88d0

2. CREATION-TOOL AND METADATA EVIDENCE

Labels are self-asserted and can be edited or forged. Use of a tool is not evidence of falsification.

[INFORMATION] Software labels found in metadata

Detected:

Info /Producer: "ReportLab PDF Library - (opensource); pyHanko 0.38.1" -> ReportLab (PDF generation library / print path)

Establishes:

The file carries self-asserted labels that are consistent with the software named above having created or processed it.

Unknown:

Labels can be edited, stripped or forged, and they do not show what was done to the content. Use of any of these tools is not evidence that the document was falsified.

[INFORMATION] Document modified after it was created

Detected:

Created 2000-01-01 00:00:00 UTC; modified 2026-11-06 19:25:29 UTC (9806 days later).

Establishes:

The metadata records at least one save after the original creation.

Unknown:

What was changed. Re-saving, signing, compressing or form-filling all update ModDate.

3. SIGNATURE INFORMATION

Signature Signature1 - Grace Banda

1. Integrity	Cryptographically valid. the signature value verifies and the digest of the signed bytes matches
2. Certificate	Chains to a trusted root. Trust set: test root CA (fixture generator)
Revocation	Status unknown. no usable embedded revocation evidence; online lookup is never performed (the app has no network access)
3. After signing	4 page content, existing images/fonts/forms, or pages; 1 overlay annotation (stamp, text box, shape); 2 new appearance streams, fonts or images (not referenced by page content on their own)
Claimed time	D:20261106192529Z (the signer's own claim unless a verified timestamp is shown)

A valid signature does not show that the signed information is true: a document made with false information and then signed still verifies.

[ESTABLISHED] Signature "Signature1": the document was changed after it was signed

Detected:

After the signed revision, 4 page content, existing images/fonts/forms, or pages change(s); 1 overlay annotation (stamp, text box, shape) change(s); 2 new appearance streams, fonts or images (not referenced by page content on their own) change(s) were appended. The signature carries no change restrictions of its own. Changed after signing: page content, existing images/fonts/forms, or pages.

Establishes:

The signature is cryptographically valid for the earlier revision, which anchors that version as what "Grace Banda" signed. The later revisions contain different content, so the file now differs from the signed version. The signed version can be reconstructed and compared.

Unknown:

Whether the later change was legitimate. Some viewers and workflows append content after signing with the signer's consent, and whether the signature's own terms permit a change is a policy question this tool answers only for DocMDP. It does not show who made the change or why.

4. EDITING AND REVISION INDICATORS

Indicators are prompts for human review, not proof of manipulation.

[INDICATOR] Revision 2: visible content was changed after the previous save

Detected:

page content changed on page 1; 1 new font; overlay-capable annotations added (Redactx1); 1 form XObject(s) changed.

Establishes:

The file's own history shows that content objects were added or replaced after revision 1. The earlier version is recoverable and can be compared with this one.

Unknown:

Whether the change was legitimate (e.g. a correction, completing a form, a stamp) or deceptive. A document carrying both versions inside it is also something an editor can construct deliberately, so this history is not an independent, trusted original.

[INDICATOR] Unreferenced objects remain in the file

Detected:

2x Page content stream (obj 1, 9); 1x Annotation (obj 14); 1x Form XObject (obj 15)

Establishes:

These objects are live in the cross-reference table but nothing in the document refers to them, so no viewer

displays them. At least some are page content, pages or images, which can mean content was removed from the document without being purged from the file.

Unknown:

Writers that do not garbage-collect leave unreferenced objects behind routinely; deleted content is one cause among several. See the recovered text preview, if any.

[INDICATOR] Part of a line of same-size text sits at a slightly different height

Detected:

1 line(s) where some text shares a line and font size with other text but its baseline is 0.25-1.5 pt off: "Received from:" vs "Joseph Mwanza" (page 1, ?0.89 pt)

Establishes:

Typeset text that is laid out together normally shares one baseline. A run placed separately, by hand or by an editor, often lands a fraction of a point off.

Unknown:

Superscripts, inline images, some layout engines and rounding in converters produce small offsets too, and a careful editor can align inserted text exactly, so the absence of this indicator proves nothing. It is a prompt to look at the line.

1 highlighted region(s) require review (section 7).

[INFORMATION] 2 incremental updates found (3 revisions)

Detected:

The file has 3 cross-reference sections chained with /Prev; each update appends new bytes after the previous %%EOF.

Establishes:

Earlier versions of this file exist inside it and can be reconstructed and compared.

Unknown:

Why the updates were made. Signing, form filling, adding annotations and compression-free re-saves all create incremental updates.

[INFORMATION] Revision 1: update without visible-content changes

Detected:

1 form field object(s) changed; 1 signature object(s) added; document information changed (2).

Establishes:

An update was appended, but none of the page content, images, fonts or page structure changed.

Unknown:

Whether the metadata, forms or annotations touched are material to the document's meaning.

[INFORMATION] 2 font face(s) used

Detected:

Helvetica-Bold - Type1 - not embedded - pages 1 | Helvetica - Type1 - not embedded - pages 1 | Helvetica - Type1 - not embedded - pages 1

Establishes:

These are the font resources the pages declare.

Unknown:

Fonts show how text was typeset, not what it says or who wrote it. Software substitutes fonts during conversion, so the same visible document can legitimately use different fonts in different files.

[INFORMATION] Page composition

Detected:

1 page(s): vector text.

Establishes:

Shows whether each page is typeset text, a scanned picture, or a picture with text on it.

Unknown:

Whether a scan faithfully shows its source. Image-only pages cannot be text-compared without OCR, which this tool does not perform.

5. FILE COMPARISON (ORIGINAL VERSUS SUBMITTED)

No trusted original, hash or issuer record was supplied. The original contents of this document are therefore unknown to this tool, and no statement about them is made.

[LIMITATION] No trusted original supplied

Detected:

No reference file, hash or issuer record was provided.

Establishes:

Nothing about the document's ORIGINAL contents is known to this tool. All findings compare the file only against itself.

Unknown:

What the document originally said. To test content, supply an authenticated original (PDF, image or .docx), a trusted hash, or a verified issuer record.

6. IMAGES AND SCANS

Image indicators are never proof of editing; each method has documented blind spots.

[INFORMATION] 1 embedded JPEG image(s) inspected

Detected:

obj 4 (page 1): 160x200, 4:2:0, baseline, standard tables ~ quality 88

Establishes:

How each JPEG was written: compression tables, sampling and metadata.

Unknown:

Tables show the LAST compression only. Non-standard tables are normal for cameras, phones and some scanners.

7. HIGHLIGHTED REGIONS REQUIRING REVIEW

* page 1 (submitted): x 72, y 164, 81 x 12 pt - baseline offset: "Received from:"

8. LIMITATIONS

* This tool reports what could be demonstrated from the supplied bytes. It cannot certify a document as genuine.
* Metadata, tool labels, XMP history and dates are self-asserted by the file and can be edited or forged.
* Without a trusted original, the document's original contents are unknown; indicators compare the file only against itself.
* A PDF that was fully regenerated (printed to PDF, converted, optimised) carries no recoverable history of earlier versions. This is a limit of the evidence, not a sign of innocence or guilt.
* A valid digital signature shows the signed bytes are unchanged since signing; it does not show that what was signed is true.
* Indicators are prompts for human review and are not accuracy-rated: none of the checks has been statistically calibrated, so no probability or confidence percentage is given.

Specific to this file

* No trusted original supplied: What the document originally said. To test content, supply an authenticated original (PDF, image or .docx), a trusted hash, or a verified issuer record.

9. REPRODUCIBILITY

Tool	Doc Forensics Engine 0.1.0
Ruleset	2026.11-r1
PDF library	com.tom-roush:pdfbox-android:2.0.27.0
Crypto library	org.bouncycastle:bcprov-jdk18on:1.83
Trust set	test root CA (fixture generator)
Checks run	file fingerprints (SHA-256/512, SHA-1, MD5); PDF metadata and creation-tool labels; revision / incremental-update history; cross-reference integrity, page structure, fonts, unreferenced objects; text, fonts and paint order (covered, hidden, overlaid content); embedded images: JPEG structure, placement over scans, noise consistency; digital signatures (byte ranges, CMS verification, certificate trust, changes after signing); comparison skipped: no trusted original supplied

The complete findings are attached to this PDF as report.json (schema doc-forensics-report/1). The examined

document itself is not included.